Genetics and Immunology in Pediatric Gastroenterology: When to Sequence, How to Screen, and Which Test to Choose.

Picture a three-month-old boy with bloody diarrhea since the second week of life. He has a perianal fistula, an abscess that keeps coming back, and a crusting rash of inflamed hair follicles on his scalp. He has already had steroids and a first dose of an anti-tumor necrosis factor antibody, and he is no better. Now picture a fifteen-year-old boy with Crohn disease that keeps forming fistulas despite two biologics, who was treated for a strange febrile illness with a very high ferritin when he was eight. And picture a nine-year-old girl admitted with her third episode of pancreatitis in a year, whose imaging is normal and whose chart says idiopathic.

All three children may have a single-gene disease, and in all three the genetic answer would change what you do next. The clinical question of this chapter is when a child with a gastrointestinal disease needs genetic testing, what immunological screening should come first, and which genetic test to order. The main idea is this. Genetic testing in pediatric gastroenterology is not a fishing trip. It is targeted by age at onset, by phenotype and by red flags; it is preceded by a cheap immunology screen; it is run as trio sequencing whenever possible, meaning the child and both parents; and its results are confirmed functionally before they drive a treatment as drastic as a bone marrow transplant. The stakes are concrete. In a large Canadian sequencing study, about one percent of all children with inflammatory bowel disease carried a variant potentially curable by stem cell transplantation, and the wrong treatment, whether the wrong biologic or a transplant for a disease a transplant cannot fix, can do real harm.

A note on sources. For monogenic inflammatory bowel disease we use the 2021 position paper of the Pediatric IBD Porto group of the European Society for Paediatric Gastroenterology, Hepatology and Nutrition, known as ESPGHAN, written by Uhlig and colleagues; the 2020 position paper on very early onset disease from the North American Society for Pediatric Gastroenterology, Hepatology and Nutrition, known as NASPGHAN, written by Kelsen and colleagues; and the 2023 consensus guideline of the British Society of Gastroenterology and the British Society of Paediatric Gastroenterology, Hepatology and Nutrition, written by Kammermeier and colleagues. The key cohorts are the Toronto exome study by Crowley and colleagues in 2020, the Boston very early onset cohort by Collen and colleagues in 2022, the French multicenter study by Charbit-Henrion and colleagues in 2018, and a systematic review of published monogenic cases by Nambu and colleagues in 2022. For congenital diarrheas we use the PediCODE consortium framework by Thiagarajah and colleagues in 2018 and the 2025 New England Journal of Medicine study by Gaibee and colleagues. For pancreatitis we use the INSPPIRE consortium papers by Kumar, Giefer and Gariepy. For polyposis we use the 2019

ESPGHAN position papers. And for choosing a test we use the 2018 meta-analysis by Clark and colleagues.

Part one. Age bands and what monogenic means.

Start with vocabulary. Very early onset inflammatory bowel disease, often shortened to VEO-IBD, means disease that begins before six years of age. Within it, infantile or toddler onset means before two years, and neonatal onset means within the first twenty-eight days of life. These bands matter because the standard Paris classification of pediatric inflammatory bowel disease groups everything from birth to just under ten years into a single category, and so does not single out the youngest children.

Monogenic inflammatory bowel disease means that a variant in a single gene is sufficient by itself to cause intestinal inflammation. Most of these genes encode parts of the immune system, so the disease is really a primary immunodeficiency that shows itself in the gut; others encode parts of the intestinal epithelium, the lining of the bowel. This is quite different from the common polygenic risk variants, such as those in NOD2, which nudge risk up a little and do not change treatment. A 2022 analysis by Bolton and colleagues counted one hundred and two genes with high or moderate penetrance for inflammatory bowel disease-like inflammation, and the ESPGHAN consensus panel lists seventy-five.

Age is a probability modifier, not a rule. Monogenic disease is most likely in the youngest children, but it has been reported in adolescents and adults, almost always with clues outside the gut.

Part two. How often sequencing finds an answer.

The most useful single number comes from Crowley and colleagues in Toronto, who performed whole exome sequencing in one thousand and five children with inflammatory bowel disease of all ages. Thirty-one, about three percent, carried a variant in a known monogenic gene; the commonest were XIAP in five children, DOCK8 in three, and FOXP3, GUCY2C and LRBA in two each. The yield depended strongly on age. It was seven point eight percent in children diagnosed before six years and two point three percent in children diagnosed between six and eighteen. A Boston cohort by Collen and colleagues found almost the same figure in very early onset disease: seventeen of two hundred and sixteen children, seven point nine percent. In a French study by Charbit-Henrion and colleagues, which deliberately enrolled children with severe very early onset disease or a suspected monogenic disorder and used targeted next-generation sequencing, the yield rose to thirty-two percent. The ESPGHAN paper summarizes yields of thirteen to forty-one percent in infantile onset disease across cohorts.

So the yield depends on who you test. Unselected testing of every child with inflammatory bowel disease finds about three in one hundred. Testing children who are both young and clinically unusual finds about one in three.

Part three. Red flags and phenotype.

In the Toronto study, four features were associated with a monogenic diagnosis. Extra-intestinal manifestations carried an odds ratio of about fifteen, onset before two years about six, a family history of autoimmunity about five, and the need for surgery about three and a half. The ESPGHAN red flags are worth memorizing as a set: onset before two years; consanguinity, an affected relative, or several relatives with early onset disease; recurrent or atypical infections, or hemophagocytic lymphohistiocytosis, a hyperinflammatory syndrome often shortened to HLH; autoimmunity, particularly the pattern of immune dysregulation seen in a disorder called IPEX; malignancy; and multiple intestinal atresias. Two physical clues deserve special mention. Skin and hair problems, such as folliculitis, eczema, recurrent abscesses or abnormal hair, point toward the interleukin ten pathway, IPEX, Wiskott-Aldrich syndrome or TTC7A deficiency. And perianal fistulas and abscesses in the first year of life point strongly to interleukin ten or interleukin ten receptor deficiency.

Where the inflammation sits also matters. In the French study, a molecular diagnosis was found in sixty-one percent of children with small bowel inflammation, thirty-nine percent of those with colitis and perianal disease, and eighteen percent of those with colitis alone. The systematic review by Nambu and colleagues, which collected seven hundred and fifty published cases of monogenic inflammatory bowel disease, adds the long view. About sixty-three percent began before six years, but about eleven percent began after eighteen. About three quarters had an extra-intestinal problem at some point, including atypical infection in about forty-five percent, skin disease in about thirty-eight percent and autoimmunity in about twenty-two percent. And almost a quarter received a hematopoietic stem cell transplant.

Part four. Genes that change treatment.

The reason to find a monogenic diagnosis is that, for several genes, the diagnosis is the prescription. Loss of interleukin ten signaling, from variants in IL10, IL10RA or IL10RB, causes severe infantile enterocolitis with perianal disease and folliculitis; biologics fail, and the original 2009 New England Journal of Medicine report by Glocker and colleagues showed that allogeneic stem cell transplantation can cure it, because the defect lies in blood-derived immune cells. Chronic granulomatous disease, caused by variants in CYBB or in autosomal recessive genes such as NCF1 and CYBA, is also curable by transplant, needs antimicrobial prophylaxis, and calls for caution with anti-tumor necrosis factor therapy because of serious infections. IPEX, from FOXP3 variants, presents as infant enteropathy with eczema and early diabetes in a boy; it is treated by transplant, with sirolimus as a bridge. LRBA deficiency and

CTLA4 haploinsufficiency cause enteropathy with autoimmune cytopenias and lymphoproliferation, and a 2015 Science paper by Lo and colleagues showed dramatic and sustained improvement with abatacept, a drug that replaces the missing CTLA4 function.

Two lessons limit enthusiasm for transplant. The first is that only defects of blood-derived cells are corrected by a new bone marrow. Epithelial defects, such as TTC7A deficiency, which causes multiple intestinal atresias with immunodeficiency, or EPCAM deficiency, which causes tufting enteropathy, are not fixed in the gut by a transplant. So every child should have a genetic diagnosis before a transplant, and an epithelial cause should be excluded before one is offered.

The second lesson is XIAP deficiency. In a 2022 international study of one hundred and sixty-seven patients by Yang and colleagues, overall survival at age sixteen was ninety percent with conservative management and fifty-three percent after transplantation. That comparison is not randomized, and the transplanted children were sicker, often with HLH or refractory bowel disease. But it means that a curable gene is not an automatic transplant. The decision belongs to an experienced center after a multidisciplinary genomic review. The British guideline adds a practical screen: in adolescent and young adult males with refractory fistulizing Crohn disease, test XIAP expression and neutrophil function.

Part five. The immunology screen.

Before or alongside sequencing, a basic immunology screen is cheap, quick and often decisive. The ESPGHAN list starts with a complete blood count with differential, looking for neutropenia, lymphopenia, or a low platelet count with small platelets, which suggests Wiskott-Aldrich syndrome. Then come immunoglobulin G, A, M and E against age-specific ranges; immunoglobulin E is often high in IPEX, DOCK8 deficiency and Wiskott-Aldrich syndrome. Then lymphocyte subsets, meaning counts of T cells, B cells and natural killer cells, with naive and memory T cells, and regulatory T cells if IPEX is suspected. And then the dihydrorhodamine test, usually called the DHR test, for chronic granulomatous disease. When a specific disorder is suspected, targeted functional tests follow: FOXP3, XIAP, LRBA and CTLA4 protein expression by flow cytometry, interleukin ten-induced phosphorylation of the signaling protein STAT3 to test the interleukin ten receptor pathway, and the monocyte response to a bacterial fragment called muramyl dipeptide to test XIAP function. Vaccine antibody responses, T cell receptor excision circles, anti-enterocyte antibodies, HIV testing and a tuberculosis interferon gamma release assay are added as indicated.

One trap deserves emphasis. A normal amount of FOXP3 or XIAP protein does not exclude the disorder, because a missense variant can produce a normal quantity of a protein that does not work.

Part six. Chronic granulomatous disease and the DHR test.

Chronic granulomatous disease is the immunodeficiency gastroenterologists most often meet, because the bowel is so often involved. In a series of one hundred and forty patients followed at the National Institutes of Health, reported by Marciano and colleagues in 2004, thirty-three percent had gastrointestinal involvement, and eighty-nine percent of those had the X-linked form. The median age at the first gastrointestinal symptom was five years, and seventy percent began in the first decade. All affected patients had abdominal pain, seventy percent had low albumin, and seventy-one percent relapsed after prednisone, so steroid dependence is common.

The DHR test works like this. Neutrophils are stimulated with a chemical called phorbol myristate acetate, which switches on the enzyme complex called NADPH oxidase. The reactive oxygen this produces converts dihydrorhodamine into fluorescent rhodamine, which flow cytometry measures cell by cell. If the signal is absent, think of X-linked disease from CYBB. If it is low but present, think of autosomal recessive disease, often from NCF1. And if there are two populations of neutrophils, one bright and one dark, you are looking at an X-linked carrier, typically the mother. Myeloperoxidase deficiency can give a falsely low result, so confirm genetically. Remember too that carrier mothers and girls with skewed X inactivation can themselves have colitis, discoid lupus and infections, so check the mother of every X-linked case.

Part seven. Congenital diarrheas and enteropathies.

The PediCODE consortium, in a 2018 paper by Thiagarajah and colleagues, reorganized the approach to diarrhea that starts in the first weeks of life. It replaced the old term intractable diarrhea of infancy with congenital diarrheas and enteropathies, abbreviated CODEs, renamed osmotic diarrhea as diet-induced, and renamed secretory diarrhea as electrolyte transport-related. The algorithm starts with the stool. The stool osmotic gap is calculated as two hundred and ninety minus twice the sum of stool sodium and potassium. A gap above one hundred milliosmoles per kilogram suggests diet-induced diarrhea, which stops with fasting, as in sugar transport and enzyme defects. A gap below fifty suggests an electrolyte transport defect, which continues when the infant is fasted, as in congenital chloride diarrhea from SLC26A3 or congenital sodium diarrhea from SLC9A3 or GUCY2C. Reducing substances above half a percent or a stool pH below five point three point to carbohydrate malabsorption. Fatty stools lead toward pancreatic insufficiency, abetalipoproteinemia and chylomicron retention disease, and bloody stools toward very early onset inflammatory bowel disease, IPEX, autoimmune enteropathy and immunodeficiency. Exome sequencing is built into the algorithm.

The 2025 study by Gaibee and colleagues shows why. Among one hundred and twenty-nine probands with congenital diarrhea and enteropathy who had exome or genome sequencing, forty-eight percent, sixty-two children, had a causal variant in a known gene. Ninety-four percent of these were autosomal recessive and six percent X-linked. The largest group,

forty-four percent of the diagnoses, were defects of epithelial trafficking and polarity, such as EPCAM, MYO5B, TTC7A and SKIV2L. The study also proposed three new genes with functional support. A few classic examples are worth knowing: MYO5B causes microvillus inclusion disease, with watery diarrhea from the first days of life that continues when fasted; EPCAM causes tufting enteropathy; DGAT1 causes vomiting, diarrhea and protein-losing enteropathy with fat in the feeds; and NEUROG3 causes enteric anendocrinosis, a malabsorption followed later by diabetes. A genetic diagnosis in these infants informs prognosis, the need for long-term parenteral nutrition, and candidacy for intestinal transplantation.

Part eight. Pancreatitis.

In children with acute recurrent or chronic pancreatitis, genes are the commonest risk factor. The INSPPIRE consortium, in a 2016 JAMA Pediatrics paper by Kumar and colleagues, studied three hundred and one children: one hundred and fifty-five with acute recurrent pancreatitis and one hundred and forty-six with chronic pancreatitis. At least one variant in a pancreatitis gene was found in forty-eight percent of the acute recurrent group and seventy-three percent of the chronic group. Variants in PRSS1, the cationic trypsinogen gene, were found in seventeen percent versus forty-six percent, with an odds ratio of about four for chronic disease; SPINK1 in thirteen versus twenty-five percent; CFTR in thirty-four versus twenty-three percent; and CTRC in ten versus five percent. Obstructive factors were present in a third of both groups, a reminder that genes and anatomy often coexist. In the 2017 analysis by Giefer and colleagues, children whose first attack came at five years or younger had a variant in seventy-one percent, with PRSS1 in forty-three percent. And a 2026 report from the same consortium found a risk variant in seventy-four percent of nine hundred and forty-four children on a fourteen-gene analysis, with any risk variant predicting faster progression from recurrent to chronic disease.

The 2017 INSPPIRE consensus, reported by Gariepy and colleagues, states that the genetic evaluation of acute recurrent or chronic pancreatitis should include a sweat chloride test and PRSS1 testing, and that SPINK1, CFTR and CTRC should be considered. Why test? A genetic cause ends the search for rarer causes, predicts progression, informs decisions about total pancreatectomy with islet autotransplantation, and allows testing of relatives. One nuance: a single CFTR variant with a borderline sweat test places a child on the CFTR-related disorder spectrum. It calls for extended CFTR analysis, not a diagnosis of cystic fibrosis.

Part nine. Polyposis.

In polyposis, a genetic result starts the surveillance clock. The 2019 ESPGHAN position papers give the pediatric schedule. In familial adenomatous polyposis, caused by APC variants, predictive testing of at-risk children is recommended at twelve to fourteen years, or

earlier if there is rectal bleeding, with colonoscopy from twelve to fourteen and then every one to three years depending on the polyp burden; upper endoscopy is not needed before twenty-five, and routine hepatoblastoma screening is not recommended. In Peutz-Jeghers syndrome, from STK11 variants, predictive testing is offered in early childhood, and upper endoscopy, colonoscopy and capsule endoscopy are done by age eight and then every three years, with elective removal of small bowel polyps larger than about one and a half to two centimeters to prevent intussusception, and testicular examination in boys for Sertoli cell tumors. In juvenile polyposis, from SMAD4 or BMPR1A variants, testing and colonoscopy begin at twelve to fifteen. A single juvenile polyp, by contrast, needs no repeat colonoscopy and no genetic test. And SMAD4 is a vascular gene too: juvenile polyposis with SMAD4 overlaps hereditary hemorrhagic telangiectasia, so these children need screening for pulmonary and cerebral arteriovenous malformations.

Part ten. Panel, exome or genome.

Which test should you order? The broadest evidence is the 2018 meta-analysis by Clark and colleagues, which pooled thirty-seven studies and more than twenty thousand children with suspected genetic disease. Genome sequencing gave a diagnosis in forty-one percent, exome sequencing in thirty-six percent, and chromosomal microarray in ten percent. Results changed management in twenty-seven percent of children with genome sequencing, seventeen percent with exome and six percent with microarray. Trio sequencing roughly doubled the odds of a diagnosis compared with testing the child alone, with an odds ratio of about two, and interpretation in a hospital close to the clinical phenotype outperformed interpretation in a distant reference laboratory, forty-two against twenty-nine percent.

For gastrointestinal indications, the platforms are complementary rather than a strict hierarchy. A targeted panel gives the deepest and most accurate coverage of known genes, detects copy number variants when designed to, and costs, according to ESPGHAN, about a quarter of an exome; but it cannot find a gene it does not contain. An exome allows a virtual panel to be analyzed first and widened later, but in the French study it missed large deletions and duplications that targeted sequencing found. A genome gives even coverage including non-coding regions and is the best substrate for reanalysis; the British guideline therefore prefers it, and the English health service offers a rapid genome for critically ill children with a mean turnaround of about nine days. ESPGHAN considers all three acceptable and recommends choosing by availability and suspicion. Whatever you choose, sequence the parents too when you can, because parental samples show whether two variants are on different copies of the gene and whether a variant is new in the child.

Part eleven. Interpreting the result.

A variant is a hypothesis until it is proven. Laboratories classify variants on the five-tier scale of the American College of Medical Genetics and Genomics: pathogenic, likely pathogenic, uncertain significance, likely benign and benign. A pathogenic or likely pathogenic variant that fits the phenotype can guide management and testing of relatives. A variant of uncertain significance should never drive treatment on its own; it calls for a functional assay and for checking whether it segregates with disease in the family. ESPGHAN recommends, with unanimous agreement of its panel, that novel genes and variants of uncertain significance be functionally confirmed before they guide therapy, and that data be shared through public databases. The danger case is obvious: a variant of uncertain significance in a gene curable by transplant must never trigger a transplant without functional proof. And a negative test is not the end. Reanalyze in one to two years, move from panel to exome to genome, and add copy number or RNA studies if suspicion remains.

Part twelve. What the guidelines say.

North America and Europe agree on the core. Both the NASPGHAN position paper and the ESPGHAN Porto group recommend evaluating infantile and very early onset disease for monogenic causes, especially with red flags, and both recommend the same basic immunology screen of blood count, immunoglobulins, lymphocyte subsets and the DHR test. They differ on platform: ESPGHAN calls panels, exomes and genomes complementary, while the British guideline prefers genome sequencing. They differ on older patients: the British guideline specifically recommends XIAP and neutrophil function testing in adolescent and young adult males with refractory fistulizing Crohn disease. For pancreatitis, the INSPPIRE consensus gives clear North American guidance on sweat chloride and PRSS1 testing, and we did not identify a separate ESPGHAN statement on genetic testing, which leaves a gap. For familial adenomatous polyposis, the 2015 American College of Gastroenterology guideline starts colonoscopy at ten to twelve years, slightly earlier than the twelve to fourteen years recommended by ESPGHAN in 2019. And on variants of uncertain significance both sides rely on the American College of Medical Genetics framework, with ESPGHAN explicitly requiring functional confirmation. The most important shared point is negative: no guideline recommends routine sequencing of every child with inflammatory bowel disease, because the yield in unselected cohorts is only about three percent.

Part thirteen. Traps.

The traps cluster in a few places. The first is treating an infant as if this were ordinary inflammatory bowel disease: giving anti-tumor necrosis factor therapy for interleukin ten receptor colitis, or performing a colectomy before a diagnosis, and losing months. The second is trusting a normal screen: a normal FOXP3 or XIAP protein level, an exome that missed a copy number variant, or a panel that simply did not include the gene. The third is acting on a variant of uncertain significance, whether by transplanting without functional proof, by

counseling a family wrongly, or by never reanalyzing the report. The fourth is the wrong cure: a transplant for an epithelial defect, a reflex transplant in XIAP deficiency, or a carrier mother with chronic granulomatous disease who is never examined. And in pancreatitis, the trap is calling a child's third attack idiopathic without a sweat test and PRSS1.

Part fourteen. What to do differently in clinic.

The practical sequence has five steps. First, define the phenotype: age at onset, location of inflammation, perianal disease, skin and hair, infections, autoimmunity and a three-generation family tree. Second, run the immunology screen: blood count, immunoglobulins, lymphocyte subsets and DHR, with targeted flow cytometry if a specific disorder is suspected. Third, sequence, ideally as a trio, choosing panel, exome or genome by availability, and asking for a rapid genome if the child is critically ill. Fourth, confirm: functional assays for uncertain variants, segregation in the family, and copy number or RNA studies if the test is negative. Fifth, decide in a genomic multidisciplinary team: transplant, abatacept, avoidance of futile surgery, testing of relatives, and surveillance.

Now return to the three children from the opening, and add three more. The three-month-old with colitis, a perianal fistula and folliculitis has the interleukin ten pathway at the top of the list: send the interleukin ten signaling assay and trio sequencing, and refer early to a transplant center. A four-year-old with granulomatous colitis and a liver abscess needs a DHR test now; if it is absent, sequence CYBB, be cautious with anti-tumor necrosis factor therapy, and test the mother. The fifteen-year-old boy with refractory fistulizing Crohn disease and a past episode of HLH needs XIAP expression and sequencing, and a genomic team review before any thought of transplant. A newborn with watery diarrhea that continues when fasted needs stool electrolytes and an osmotic gap, and then trio exome for congenital diarrhea genes. The nine-year-old with her third attack of pancreatitis needs a sweat chloride and testing of PRSS1, SPINK1, CFTR and CTRC, along with imaging for anatomy. And a twelve-year-old whose father has familial adenomatous polyposis needs predictive APC testing with counseling, and colonoscopy if the result is positive.

The single sentence to carry away is this: test the young and the unusual, screen immunity before you sequence, sequence the family not just the child, and let only a confirmed variant change the treatment.